

Healthcare Billing Fundamentals

What Is Hospital Billing?

Revenue Cycle Intro

A hospital bill is the end product of a 7-step pipeline called the revenue cycle. Understanding each step helps explain why a bill looks the way it does, why charges appear months after a visit, and why patients receive multiple documents from different senders.

Step	What Happens	Who Is Involved
1. Patient Registration	The hospital collects insurance details, demographics, and authorization before or during the visit.	Front desk / patient access team
2. Charge Capture	Every service, supply, and medication is recorded using billing codes (CPT, HCPCS, revenue codes).	Clinical staff + billing system
3. Coding	Medical coders translate clinical documentation into standardized codes and package them into a claim form (UB-04 or CMS-1500).	Coding / billing department
4. Claim Submission	The coded claim is sent electronically to the payer (insurer, Medicare, Medicaid).	Billing department
5. Adjudication	The payer applies contract terms, patient benefits, and deductibles to determine the allowed amount and patient share.	Insurance company
6. Remittance / EOB*	Payer sends an Explanation of Benefits (EOB) to both the patient and the provider showing what was paid and why.	Insurance company
7. Patient Billing*	The provider sends the patient a statement for any remaining balance after insurance is paid.	Hospital billing department

***Our agent operates at Steps 6-7 (the patient already has a statement or EOB in hand and does not understand it). Every section below maps to knowledge the agent needs to explain that document.**

Key Billing Code Systems

Every line item on a hospital bill is encoded using one or more of four code systems. The explain_bill tool must translate any code a patient sees into plain language.

Code System	What It Codes	Example	Appears On	Agent Relevance
CPT (Current Procedural Terminology)	- Procedures and services performed - AKA what was done to the patient.	47562 = Laparoscopic cholecystectomy 99213 = Office visit, established patient	- UB-04 - CMS-1500	- Most common code patients see. - Must be translated to a plain procedure name and context.
ICD-10-CM (Diagnosis Codes)	- Diagnoses (why the service was needed). - Also drives medical necessity review.	- K80.20 = Gallstones - I10 = High blood pressure - Z00.00 = Annual checkup	- UB-04 - CMS-1500	- Patients fear their diagnosis code. - Agent must give billing context without providing clinical interpretation ((always redirect clinical questions to their physician)).

HCPCS Level II (Supplies, Drugs, Equipment)	- Items CPT does not cover. - Injectable drugs, medical supplies, ambulances, durable equipment.	- J0696 = Ceftriaxone antibiotic injection - A6216 = Gauze dressing - A0425 = Ambulance mileage	- UB-04 - CMS-1500	- Often appear as cryptic line items. - Agent must identify whether it is a drug, supply, or service and explain it in plain language.
Revenue Codes (UB-04 only)	- Hospital department where a service was provided. - Used only on facility (UB-04) bills.	0360 = Operating Room 0272 = Sterile Supplies 0730 = EKG/ECG Lab	- UB-04 only - NEVER on CMS-1500	- Most confusing code type for patients. - Agent must translate revenue codes to department names.
DRG (Diagnosis Related Group)	- Inpatient payment bundle. - Medicare and many insurers pay one flat rate for the entire admission instead of billing each service.	- DRG 291 = Heart failure and shock with MCC - DRG 470 = Total joint replacement without MCC	Inpatient UB-04 only	- Patient expects to see individual line items but sees one large bundled charge. - Agent must explain why inpatient billing is different from outpatient.
Modifiers	2-character additions to CPT codes that change how a service is billed - Bilateral, separate service, professional component only, etc.	-25 = Significant separate E&M same day -50 = Bilateral procedure -TC / -26 = Technical / professional component	- UB-04 - CMS-1500	- Patients read modifiers as duplicate charges. - Agent must identify which modifier applies and explain why it is not a billing error.

Key Billing Documents

A single hospital visit can generate up to four separate documents from different senders. Before the agent explains anything, it must correctly identify which document it is looking at as the explanation strategy is different for each one.

Document	Sent By	When	Key Fields	Why Patients Get Confused
UB-04 (Facility Bill)	Hospital or facility	After insurance adjudication	Revenue codes, CPT codes, ICD-10 diagnoses, admission/discharge dates, total charges, insurance payment, patient balance	- Revenue codes are meaningless without translation. - Gross charges look enormous because the contractual adjustment has not been applied yet on some statement versions.
CMS-1500 (Physician Bill)	Physician, surgeon, anesthesiologist or specialist	After insurance adjudication	CPT codes, ICD-10 diagnoses, place of service, rendering provider NPI, allowed amount, patient copay/coinsurance	- Patients receive a hospital UB-04 AND a separate physician CMS-1500 for the same visit and assume they are being double-billed. - This is normal and expected.
EOB (Explanation of Benefits)	Insurance company	After insurer processes the claim	Billed amount, allowed amount, insurance paid, patient responsibility, denial codes	- The EOB says 'This is NOT a bill' but shows a patient responsibility amount. - Most patients treat it as a bill and pay it before the actual provider statement arrives or panic before any bill exists.
Patient Statement	Hospital billing department	30-60 days after service, after insurance pays	Account balance, due date, services summary, payment options, financial assistance mention	- May combine multiple visits. - Balance may differ from the EOB if payments were posted after the EOB was generated. - This gap between EOB and statement is a major source of confusion.

How a Bill Is Calculated: From Chargemaster to Patient Balance

Every patient balance is the result of a chain of adjustments starting from the hospital's list price. The agent must walk patients through this math clearly as it is one of the most common sources of confusion.

Term	Plain Meaning	Dollar Example	Agent Task
Gross / Total Charge (Chargemaster price)	The hospital's list price. Almost no one pays this. It exists so hospitals have room to negotiate discounts with insurers.	\$12,000	- Patient is shocked by the large number. - Explain this is the starting point before any discounts (they will not owe this).
Contractual / Negotiated Adjustment	The discount the hospital agreed to give the insurer per their contract. This is a write-off, NOT a payment.	- \$4,500	- Patients confuse adjustment with insurance payment. - Clarify: 'This \$4,500 is a discount built into your insurance contract, it is not money anyone paid.'

Allowed Amount	- Gross charge minus the adjustment. - The true cost under the insurance contract. - This is what insurance and patient together owe.	\$7,500	Explain this is the 'real' cost used to split responsibility between insurer and patient.
Insurance Payment	What the insurer paid directly to the hospital after adjudication.	\$6,750 (90%)	- Patient asks why insurance didn't pay the whole bill. - Explain the coinsurance split and whether any deductible applied.
Patient Responsibility	- Deductible + coinsurance + copay. - The patient's contractual share of the allowed amount.	\$750 (10%)	- This is the number on the patient's bill. - Explain which component (deductible, coinsurance, or copay) generated it.
Amount Paid to Date	Payments the patient has already made toward this balance.	\$200	Verify with patient if any payments have been made that may not yet be posted.
Balance Due*	What the patient still owes right now.	\$550	- The only number that drives the patient's next action. - Always confirm this is the final number after all adjustments.

***Note:** in many cases, balance due can be approximated as: Gross Charge – Adjustments – Insurance Payments – Patient Payments. Actual balances may also reflect secondary insurance, financial assistance, or other account adjustments

Common Denial Codes and What They Mean for Patients

When insurance partially or fully denies a claim, they use CARC codes (Claim Adjustment Reason Codes) on the EOB. Patients see these codes with no explanation. The agent should translate the most common ones and always provide a concrete next step.

CARC Code	Official Reason	Plain-Language Translation	Patient's Next Step
1	Deductible amount	- Your deductible has not been fully met yet. - You owe this portion before insurance begins paying.	- Confirm your remaining deductible with your insurer. - Ask if any other claims this year have been applied.
2	Coinsurance amount	- This is your percentage share of the allowed amount. - Insurance paid its part, and this is yours.	Verify the coinsurance percentage matches your plan documents.
96	The service is not covered	Your insurance plan does not cover this type of service under your current benefits.	- Ask your provider whether an alternative billing code applies. - Ask your insurer for the denial in writing and review appeal options.
16	Claim/service lacks information needed for adjudication	The claim was missing required information and the insurer cannot process it as submitted.	Contact your provider and ask them to resubmit with the missing documentation.

50	Non-covered service (not medically necessary)	The insurer determined this service was not medically required based on the diagnosis and clinical information submitted.	- Request the denial reason in writing. - Ask your provider to submit clinical notes supporting medical necessity. - File an appeal with supporting documentation.
97	Payment included in allowance for another service (bundled)	This charge is already included in the payment for another procedure billed on the same claim.	- This may be a correct bundling rule. - If unexpected, request an itemized bill review and ask your provider whether the code combination is correct.
45	Charge exceeds fee schedule / maximum allowable	- The hospital billed more than your insurer allows for this service. - The difference is a standard contractual write-off.	- No action needed in most cases. - If you are being billed for the excess amount by an in-network provider, that is not permitted (contact your insurer).

Top Things That Confuse Patients About Their Bills

These are the recurring confusion points that drive patients to call billing departments, and that our agent should address proactively. Each one maps directly to the billing knowledge from the last sections.

#	What Confuses Patients	Why It Happens
1	Receiving multiple bills for one visit	- Hospital (UB-04), physician (CMS-1500), anesthesiologist, lab, and radiology each bill separately. - One ER visit can generate 4–5 separate bills from different entities. - This is normal but patients have no way to know that.
2	EOB vs. actual bill	- The EOB says 'This is NOT a bill' but shows a patient responsibility amount. - Most patients either treat it as a bill and pay early, or panic before the actual bill even arrives.
3	Gross charge vs. what they actually owe	- An \$80,000 surgery resulting in a \$1,200 patient balance is confusing. - Patients do not understand contractual adjustments, so the large gross charge number seems like their debt.
4	Revenue codes on UB-04	- Codes like 0360, 0272, and 0730 mean nothing to a patient without a reference. - The UB-04 uses department codes instead of plain service names.
5	Observation status instead of inpatient admission	- A 3-night hospital stay billed as outpatient 'observation' means Medicare Part A does not apply. - The patient owes 20% of every service with no cap, and the stay does not count toward SNF eligibility.
6	Modifier on a CPT code looks like a duplicate	Seeing -25 (separate office visit same day as a procedure) or -50 (bilateral procedure) attached to a code looks like the same service was charged twice.
7	Insurance paid less than expected	Patient expected 80% coverage but insurance paid 60% because the deductible was not fully met, or because coinsurance applies after the deductible.
8	Contractual adjustment labeled as 'write-off' or 'discount'	Patients think this is something they personally negotiated, when it is a pre-set discount from the insurer's contract and they cannot claim it as uninsured patients without the same contract.

9	Preventive care copay surprise	• ACA-mandated preventive screenings should cost \$0. • However, if a diagnostic code is added or a therapeutic action is taken during the visit (e.g., polyp removal during colonoscopy), some insurers reclassify the visit and apply cost-sharing, even though CMS guidance says this should not happen for Medicare screenings. • Patients should verify and appeal if charged.
10	Prior balance on statement	• A statement may combine current and past visit balances. • Patients cannot tell which visit generated which charge, especially when account numbers differ.

What This Means for the explain_bill Tool

The sections above define what the explain_bill tool needs to know. Below maps each area of billing knowledge to a specific agent task and identifies the failure mode that would result if that knowledge is missing.

Agent Task	Billing Knowledge Required	Failure Mode to Avoid
Identify document type before explaining	• Distinguish UB-04, CMS-1500, EOB, and patient statement by their fields and sender.	• Treating an EOB as a bill. • Telling a patient they owe money when payment has not yet been requested by the provider.
Translate every billing code	• CPT → procedure description • ICD-10 → plain diagnosis context • HCPCS → supply or drug name • Revenue Code → department name • DRG → bundled inpatient explanation	• Leaving any code unexplained. • Hallucinating a code's meaning. • Providing a clinical interpretation of a diagnosis code instead of a billing explanation.
Walk through the financial math	• Gross charge – contractual adjustment – insurance payment – prior payments = balance due.	• Confusing the adjustment with a payment. • Telling the patient a different amount than what appears on their bill. • Treating gross charge as patient responsibility.
Flag potential billing anomalies	• Recognizing duplicate codes, modifier misuse, observation vs. inpatient, unbundled charges, and charges that don't match the revenue code.	• Missing a duplicate charge or reassuring a patient that an anomaly is normal without flagging it for verification.
Connect to financial assistance (hand off to Track 2 logic)	• Recognize when a high patient balance or uninsured/underinsured status should trigger FPL calculation and FAP recommendation.	• Fully explaining the bill without mentioning financial assistance options for a patient with a large balance and apparent financial need.
Acknowledge the limits of available knowledge	• Know which codes are in the chargemaster and knowledge docs vs. which are not.	• Hallucinating an explanation for an unknown code instead of saying 'I cannot verify this code, please call billing for a plain-language description.'

Financial Assistance